

ECON 1101 Problem Set 1

Module 2: Gains from Trade

Name:

Student ID:

Worked with:

Problem Set 1 Information

- **Due Date:** **Tue, Sep 16 at 9:00 am** (submit on Gradescope).
- **Points:** 40 (Part 1: 8 points, Part 2: 32 points)
- **Purpose:** This problem set will help you practice key course skills, including explaining economic ideas using graphs, models, and equations.
- **Study Tips:**
 1. Start by reviewing your notes. Use the textbook if you want extra explanation.
 2. Try the problem set (PS) on your own first, even if you're not sure about every question.
 3. Meet with classmates to talk through the PS and compare ideas.
 4. After discussion, write up your own answers in your own words and submit them.
- **Checklist Before Starting:**
 - ☐ I understand that AI tools should not be used while working on problem sets.
- **Checklist Before Submitting:**
 - ☐ My answers are written in my own words.
 - ☐ (If collaborated) I included the names of the classmates I worked with.
 - ☐ I saved my file as `lastname_firstname_Econ1101_PS#.pdf`, for example `Park_Seeun_Econ1101A_PS1.pdf`.

Do your best and keep it clear and concise!

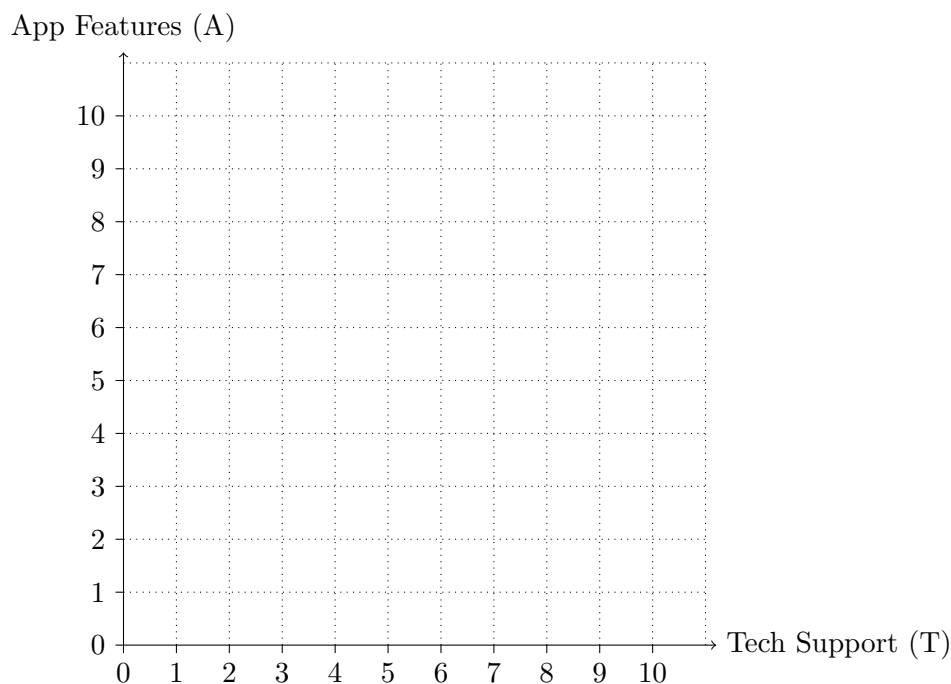
1 Bowed-Out PPF (Topic 1)

Let's revisit the tech company example from Section 1 and make it more realistic.

In reality, workers have different strengths. Some are flexible and can handle both tasks well, while others are more specialized. For example, some excel at developing apps, whereas others are especially skilled at communicating with customers and solving problems.

- (a) Plot the following production points and label them on a graph. Then connect the points smoothly to form a bowed-out (concave) PPF. No need to calculate intercepts. For a visual example, see [OpenStax 2.2](#).

- **Point A:** Your team currently includes 4 developers and 2 support staff. This setup produces 8 app features and 2 units of tech support.
- **Point B:** You reassign an intern developer to customer service. Tech support increases to 3 units, but you lose the chance to add 1 app feature. → New point: 7 app features, 3 tech support.
- **Point C:** Next, you reassign a junior developer. Tech support increases to 4 units, but you give up the chance to add 2 more app features. → New point: 5 app features, 4 tech support.
- **Point D:** Finally, you reassign your senior developer. Tech support increases to 5 units, but you give up the chance to add 4 more app features. → New point: 1 app feature, 5 tech support.



- (b) How does the **opportunity cost of 1 tech support case** change as more developers are reassigned from app development to support roles? Does the cost increase, decrease, or remain constant? Justify your answer by explicitly comparing the opportunity costs when moving from point A to B, B to C, and C to D.
- (c) Why does the PPF in this scenario have a bowed-out (concave) shape? What does this imply about the productivity of workers as more developers are reassigned to tech support?

2 Specialization and Trade (Topic 2)

Two small tech companies—**Bears Tech** and **PolarSoft**—are deciding how to allocate their workday between two tasks: App Development and Tech Support. The table below shows what each company can produce in one full workday:

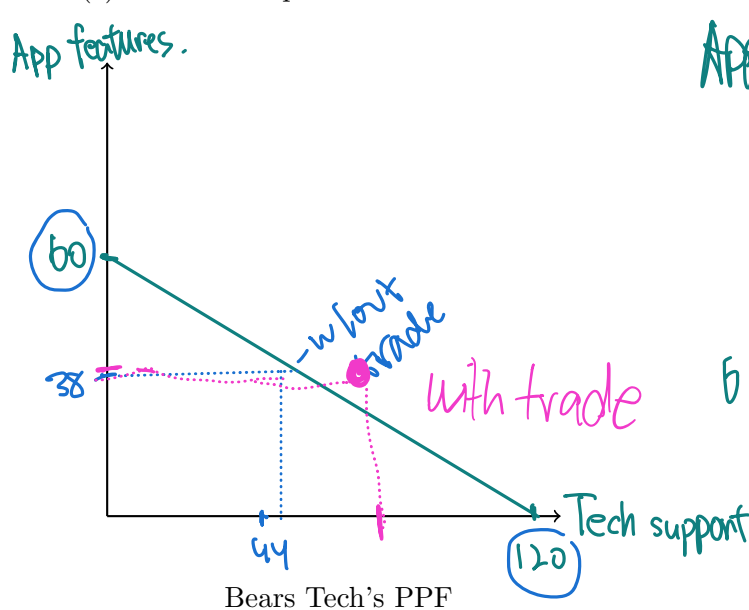
2.1 Maximum Production and the PPF

Table 1: Maximum Production

Task	Bears Tech	PolarSoft
App Features (A)	60	61
Tech Support Cases (T)	120	305

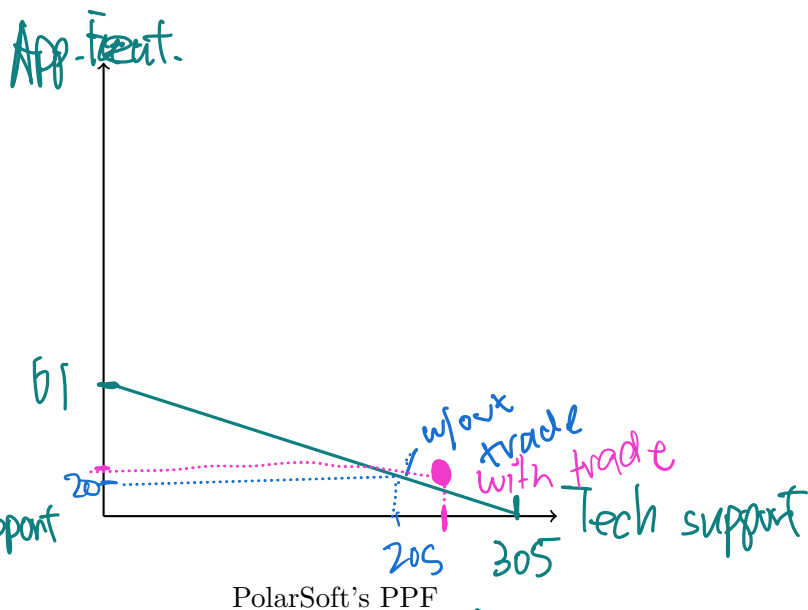
Use the table to answer the questions:

- Which company can produce more app features in a day? *PolarSoft*
- Which company can handle more tech support cases in a day? *PolarSoft*
- Which company has the **absolute advantage** in each task?
PolarSoft had an absolute advantage in both task
- Draw each firm's PPF. Place Tech Support (T) on the x -axis and App Features (A) on the y -axis. Label both axes clearly.
- Write the equation of each PPF.



$$y = a + bx$$

$$y = 60 - \frac{1}{2}x$$



$$y = 61 - \frac{1}{5}x$$

$$A = 61 - \frac{1}{5}T \quad \frac{61}{305}$$

2.2 No Trade Scenario

Table 2: Production Without Trade

Task	Bears Tech	PolarSoft	Total (Economy)
App Features	38	20	58
Tech Support Cases	44	205	249

- (a) Suppose Bears Tech develops 38 app features and PolarSoft develops 20 app features in a day. Find the resulting production point and fill in the table above. (Show your work.)

$$\begin{array}{r}
 38 = 60 - \frac{1}{2}T \\
 -60 \\
 \hline
 -22 = -\frac{1}{2}T \\
 \hline
 = 44
 \end{array}$$

$$\begin{array}{r}
 20 = 61 - \frac{1}{5}T \\
 -61 \quad -61 \\
 \hline
 -41 = -\frac{1}{5}T \\
 \hline
 -\frac{1}{5} \quad -\frac{1}{5} \\
 \hline
 = 205
 \end{array}$$

- (b) On each company's PPF, plot the production points you found in part (a) and label them as *without trade*.

2.3 Comparative Advantage and Specialization

Both CEOs turned out to be Bowdoin alumni. Remembering what they learned in Econ 1101, they decided to consider specializing in one task and trading.

Table 3: Opportunity Costs and Specialization

	Bears Tech	PolarSoft	Who Specializes?
<u>Opp. Cost of 1 App Feature</u>	2 tsc	5 tsc	Bear
<u>Opp. Cost of 1 Tech Support Case</u>	$\frac{1}{2}$ app	$\frac{1}{5}$ app	PolarSoft

(a) Fill in the opportunity cost table above.

(b) In economic terms, Bear has a **comparative advantage** in developing app features, while Polar has a **comparative advantage** in handling tech support cases.

(c) If they specialize, which firm will take care of customer support?

Pol

(d) Calculate how many app features and how much tech support each firm produces under full specialization. Fill in the production table (Table 4) below.

(e) Explain how this trade benefits the economy.

economy's total production of both goods increases

Table 4: Production With Trade

Task	Bears Tech	PolarSoft	Total (Economy)
App Features	60	0	60
Tech Support Cases	0	305	305

58 → 60

244 → 305

2.4 Terms of Trade and Gains from Trade

Now let's see how trade can make both firms better off. We focus on the **market for tech support cases (T)**.

- (a) Who is the seller of tech support?

Polarsoft!

- (b) How many app features would the seller want in exchange for 1 tech support case?

$\frac{A}{T} = \frac{1}{5}$, $\frac{1}{5}$ of an app dev for 1 tech support case.

- (c) Who is the buyer of tech support?

Bears Tech.

- (d) How many app features is the buyer willing to give up for 1 support case?

$\frac{A}{T} = \frac{1}{2}$, $\frac{1}{2}$ App dev per 1 tech support case.

- (e) Identify the range of app features per tech support (A/T) that makes trade possible.

seller AS buyer BT
 $\frac{1}{5} < TR < \frac{1}{2}$

- (f) Suppose the proposed trade ratio is 7 app features for 10 tech support cases. Which side would accept this offer and which would not?

Polar Soft would accept the $\frac{1}{5} < \frac{7}{10} < \frac{1}{2}$ deal while Bears tech. wouldn't.

- (g) The two firms agree to trade at a rate of 3 app features for 10 tech support cases. Suppose the buyer wants to acquire 70 tech support cases from the seller. After this trade takes place, how many app features and how many tech support cases will each firm be able to provide to its own consumers?

- (i) Use Table 5 to answer the question.
 (ii) Mark the point on the PPF graphs and label it as *with trade*.

$$\frac{34}{101} = \frac{214}{705}$$

Table 5: Total Services Provided With Trade

	Bears Tech	PolarSoft
App Features	$60 - 21 = 39$	$0 + 21 = 21$
Tech Support Cases	$0 + 70 = 70$	$305 - 70 = 235$

$$= 60$$

$$= 305$$

- (h) Compare the total number of app features and tech support cases each firm is able to consume (and thus provide to customers) with and without trade. Then explain how voluntary trade benefits each individual firm.

You're not done yet. Turn the page and let's wrap it up.

- (i) While trade increases total output, not everyone benefits equally. Based on our example, who gains and who loses from specialization and trade? *Keep it short. Just a couple of sentences is plenty!*
- (j) Reflecting on the example you analyzed, briefly discuss the statement: *‘There is nothing to gain from trade because we have an absolute advantage in producing both goods.’* (Hint: Use the concept of comparative advantage.)

